

Linear Regression Model for California Housing Price Prediction

1. Introduction

This project aims to build and evaluate a Machine Learning model capable of predicting house prices using the California Housing Dataset. Linear Regression was selected as the baseline algorithm because it is simple, interpretable, and widely used for continuous-value prediction tasks. The project follows the complete Machine Learning workflow including data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and recommendations for improvement.

2. Dataset Overview

The California Housing Dataset is provided by Scikit-Learn and contains housing information collected from California census districts. The target variable is `MedHouseVal`, which represents the median house value for a district. Key features include:

- `MedInc` – Median income
- `HouseAge` – Median age of houses
- `AveRooms` – Average number of rooms
- `AveBedrms` – Average number of bedrooms
- `Population` – District population
- `AveOccup` – Average occupancy
- `Latitude` and `Longitude` – Geographic coordinates

3. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset structure, identify trends, and detect potential issues. Findings:

- The dataset contains more than 20,000 records with eight input features.
- No significant missing values are present.
- Median Income (`MedInc`) shows a strong positive relationship with house prices.
- Geographic features (`Latitude` and `Longitude`) significantly influence house values.
- Some features contain outliers, especially `Population` and `AveOccup`.
- Histograms indicate that several variables are positively skewed.
- Correlation analysis generally shows that income-related features contribute most to price prediction.

EDA helps ensure that the model is trained on a well-understood dataset and highlights areas where feature engineering may improve performance.

4. Model Development

The dataset was divided into training and testing sets using an 80:20 split. The Linear Regression algorithm was trained using the training dataset. Workflow:

1. Load dataset using Scikit-Learn.
2. Separate features (`X`) and target variable (`y`).
3. Split data into training and testing sets.
4. Train `LinearRegression` model.
5. Generate predictions on unseen test data.
6. Evaluate performance using `MAE`, `RMSE`, and `R2` metrics.

Linear Regression attempts to model the relationship between input variables and house prices by fitting the best possible linear equation.

5. Evaluation Metrics

Three standard regression metrics were used: Mean Absolute Error (MAE): Measures the average absolute difference between predicted and actual values. Lower MAE indicates better accuracy. Root Mean Squared Error (RMSE): Measures prediction error while giving greater importance to large errors. Lower RMSE indicates better model performance. R^2 Score (Coefficient of Determination): Measures how much variance in house prices is explained by the model. Values closer to 1 indicate better predictive power. Typical results for Linear Regression on the California Housing dataset are: • MAE $\approx$ 0.53 • RMSE $\approx$ 0.74 • $R^2 \approx$ 0.58 – 0.61 Actual values may vary slightly depending on train-test split and preprocessing.

6. Results and Discussion

The Linear Regression model provides a strong baseline for house price prediction. The model successfully captures major relationships between housing features and property values. Strengths: • Fast training and prediction. • Easy to interpret coefficients. • Good baseline performance. Limitations: • Assumes a linear relationship between features and target. • Sensitive to outliers. • Cannot capture complex non-linear interactions. Although the model performs reasonably well, more advanced algorithms can often achieve significantly higher accuracy.

7. Improvement Ideas

Several techniques can improve model performance: 1. Feature Engineering - Create new features from existing variables. - Use interaction terms and ratios. 2. Data Scaling and Transformation - Normalize skewed features. - Apply logarithmic transformations. 3. Outlier Treatment - Detect and remove extreme values. - Use robust regression methods. 4. Cross-Validation - Reduce overfitting. - Improve model reliability. 5. Advanced Algorithms - Random Forest Regressor - Gradient Boosting - XGBoost - LightGBM These methods can capture non-linear relationships and often outperform Linear Regression.

8. Conclusion

This project demonstrated the complete Machine Learning lifecycle using the California Housing Dataset. A Linear Regression model was developed and evaluated using standard regression metrics. The model achieved reasonable predictive performance and provided valuable insights into factors affecting housing prices. Future work should focus on feature engineering, cross-validation, and advanced ensemble methods to further improve prediction accuracy.